

Twisted-Convolution Identities in Dimensions Four and Five from Shintani Ray Units

Draft for review

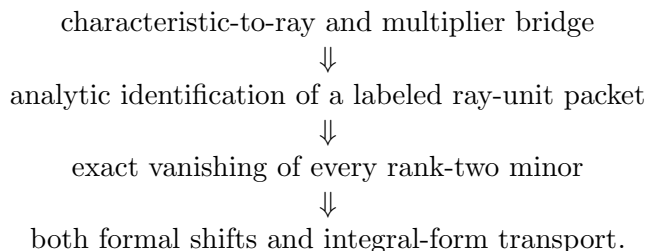
28 July 2026

Abstract

We prove the formal Twisted Convolution Conjecture unconditionally in dimensions four and five. Dimension four is the quadratic calibration: the principal-ghost table depends on one positive double-sine value, all 36 rank-two minors have one common quadratic factor, and Kopp’s Kronecker-limit formula together with an analytic class-number calculation evaluates that factor exactly. Dimension five is the main arithmetic case. For $K = \mathbb{Q}(\sqrt{3})$ and modulus $(5)\infty_2$, we compute all twenty-four nonzero characteristics, their positive Kopp lifts, ray classes, signs, and multipliers. Shintani’s 1978 algebraicity theorem gives a safe exponent 5760; labeled ray-field and Frobenius certificates, rigorous Arb enclosures, and Voutier’s height bound then identify the complete reciprocal degree-eight ray packet. Exact number-field arithmetic makes all 100 rank-two minors vanish. In both dimensions complex conjugation supplies the paired shift, and integral form covariance gives the result for every admissible tuple. No Stark conjecture or `bnrstark` recognition is used as an input. We finish with exact research-status boundaries for dimensions six and eight: one primitive order-six Shintani–Stark value remains in dimension six, while two oriented cyclic-quartic resolvents remain in dimension eight. The finite cancellation certificates in those dimensions are already exact, but the boundary results are not asserted as unconditional TCC theorems.

1 Overview and common convention

The principal-ghost construction of Appleby–Flammia–Kopp [2] associates convention-sensitive Shintani–Faddeev real-multiplication values to a finite Weyl reconstruction. The formal TCC asks for two shifted twisted convolutions to vanish, equivalently for the reconstructed ghost matrices to be rank-one idempotents. Our proof architecture is



To remove the standard double-sine ambiguity once and for all, define

$$S_2(z \mid \omega_1, \omega_2) := \frac{\Gamma_2(z \mid \omega_1, \omega_2)}{\Gamma_2(\omega_1 + \omega_2 - z \mid \omega_1, \omega_2)}. \quad (1)$$

Thus S_2 is the reciprocal of Kopp’s $\text{Sin}_{2, \text{Kopp}}$ convention [1, Definition 4.22, equation (4.8)].

Theorem 1 (Low-dimensional TCC). *For every admissible tuple in dimension four or five,*

$$0, 1 \in \mathcal{Z}_t.$$

If t, t' have the same dimension, then $\mathcal{Z}_t = \mathcal{Z}_{t'}$.

The dimension-four proof below is deliberately compact: it records the complete analytic and finite bridge but leaves the thirty-six explicit minor quotients to the supplementary certificate. Dimension five requires the longer powered-algebraicity and height-rigidity argument that forms the body of the paper.

2 Dimension four: the quadratic calibration

Put

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad \beta_4 = \phi^2, \quad Q_4 = \langle 1, -3, 1 \rangle, \quad \tau_4 = -e^{\pi i/4}.$$

The complete signed reconstruction table is

$$T_4(x) = \begin{pmatrix} \sqrt{5} & -x & 1 & -x^{-1} \\ -x^{-1} & -x^{-1} & -x^{-1} & -x \\ 1 & -x^{-1} & 1 & x \\ -x & -x^{-1} & x & -x \end{pmatrix}, \quad (\text{D4.1})$$

where

$$x = \sqrt{2} \frac{S_2(\beta_4/4 \mid \beta_4, 1) S_2(1/4 \mid \beta_4, 1)}{S_2((\beta_4 + 1)/4 \mid \beta_4, 1)}. \quad (\text{D4.2})$$

Lemma 2 (Dimension-four reconstruction table). *For the canonical tuple, the fifteen nonzero AFK analytic overlaps are the nonzero-characteristic entries of (D4.1). The entry $(T_4)_{0,0} = \sqrt{5}$ packages the normalized ghost identity coefficient into the same table.*

Proof. Put

$$\mathcal{D}_{c,b}^{(4)} := S_2\left(\frac{c + b\beta_4}{4} \mid \beta_4, 1\right).$$

For $(p, q) \neq (0, 0)$ and $r \equiv -p - q \pmod{4}$, the principal-form specialization of AFK Definition 1.32 is

$$\tilde{\nu}_{p,q} = (-1)^{4(p+q)+pq+\min(4,p+q)} \mathcal{D}_{4-p,q}^{(4)} \mathcal{D}_{4-r,p}^{(4)} \mathcal{D}_{4-q,r}^{(4)}. \quad (\text{D4.DS})$$

This formula already includes the AFK phase. The complete substitution audit is

(p, q)	r	$\tilde{\nu}_{p,q}$	(p, q)	r	$\tilde{\nu}_{p,q}$
$(0, 1)$	3	$-x$	$(0, 2)$	2	1
$(0, 3)$	1	$-x^{-1}$	$(1, 0)$	3	$-x^{-1}$
$(1, 1)$	2	$-x^{-1}$	$(1, 2)$	1	$-x^{-1}$
$(1, 3)$	0	$-x$	$(2, 0)$	2	1
$(2, 1)$	1	$-x^{-1}$	$(2, 2)$	0	1
$(2, 3)$	3	x	$(3, 0)$	1	$-x$
$(3, 1)$	0	$-x^{-1}$	$(3, 2)$	3	x
$(3, 3)$	2	$-x$			

Every nonzero-characteristic entry follows from (D4.DS) using only

$$S_2(z) S_2(\beta_4 + 1 - z) = 1, \quad S_2(z + 1) = 2 \sin(\pi z / \beta_4) S_2(z), \quad S_2(z + \beta_4) = 2 \sin(\pi z) S_2(z), \quad (\text{D4.FE})$$

and $\beta_4^2 = 3\beta_4 - 1$. The value $\sqrt{5}$ at $(0, 0)$ is not an analytic zero-characteristic cocycle value: after division by $\sqrt{5}$ it represents the Weyl coefficient $\mu_0 = 1$. $\square$

Define

$$K_{r,c}^{(4)}(x) = \frac{1}{4\sqrt{5}} \sum_{\substack{0 \leq a, b < 4 \\ r \equiv c+a \pmod{4}}} (T_4)_{a,b}(x) \tau_4^{ab} i^{bc}. \quad (\text{D4.3})$$

The representative reduction in (D4.3) retains the even-dimensional wrap sign. Its diagonal Fourier sum gives

$$\mathrm{Tr} K^{(4)} = (T_4)_{0,0}/\sqrt{5} = 1. \quad (\text{D4.4})$$

Exact Laurent reduction proves that every one of its 36 rank-two minors belongs to the principal ideal

$$\left(x^2 - \sqrt{3 + \sqrt{5}}x + 1\right) \subset \mathbb{Q}(i, \sqrt{2}, \sqrt{5})[x, x^{-1}]. \quad (\text{D4.5})$$

The certificate records all 36 quotients coefficientwise in the genuine basis $1, \sqrt{2}, \sqrt{5}, \sqrt{10}$; two minors are structurally zero and the other 34 have nonzero formal quotients.

It remains to evaluate (D4.2). Write $K_4 = \mathbb{Q}(\sqrt{5})$, $\mathcal{O}_{K_4} = \mathbb{Z}[\phi]$, and label its embeddings by $\infty_1(\sqrt{5}) = +\sqrt{5}$ and $\infty_2(\sqrt{5}) = -\sqrt{5}$. Since 2 is inert, $|(\mathcal{O}_{K_4}/4)^\times| = 12$. The images of $\{\pm\phi^n\}$ together with the sign at ∞_2 have order 12; the ray exact sequence and $h_{K_4} = 1$ therefore give

$$\mathrm{Cl}_{(4)\infty_2}(K_4) \cong C_2. \quad (\text{D4.6})$$

The field

$$L_4 = K_4(\sqrt{\phi})$$

has relative discriminant (4), and ∞_2 ramifies because $\phi' < 0$. The quadratic conductor–discriminant theorem and (D4.6) show that L_4 is the full ray field.

Set $u = \phi + \sqrt{\phi}$ and let $\sigma(\sqrt{\phi}) = -\sqrt{\phi}$. Then $u^\sigma = u^{-1}$ and

$$N_{L_4/K_4}(u) = 1, \quad \mathrm{Tr}_{L_4/K_4}(u) = 1 + \sqrt{5}.$$

The field L_4 has signature $(2, 1)$ and discriminant -400 . For $s = \sqrt{\phi}$, the power-basis discriminant of $1, s, s^2, s^3$ is also -400 , equal to the certified field discriminant; hence $\mathcal{O}_{L_4} = \mathbb{Z}[s]$. Its Minkowski bound is $15/(2\pi) < 2.4$, and there is no ideal of norm two because

$$X^4 - X^2 - 1 \equiv (X^2 + X + 1)^2 \pmod{2}.$$

Thus $h_{L_4} = 1$. PARI/GP 2.15.4 [8] returns the fundamental units $s^3 - s = s^{-1}$ and $-s^2 + s = -u^{-1}$; its `bnfcertify=1` output certifies unconditionally that

$$R_{L_4} = \log \phi \log u.$$

For the nontrivial quadratic ray character χ , $L(s, \chi) = \zeta_{L_4}(s)/\zeta_{K_4}(s)$. Using

$$\zeta_F(s) = -\frac{h_F R_F}{w_F} s^{r_1+r_2-1} + O(s^{r_1+r_2})$$

for $F = K_4, L_4$ gives

$$L'(0, \chi) = \log u. \quad (\text{D4.7})$$

We now spell out the convention-sensitive specialization of Kopp's Theorem 1.1 [1]. Take modulus $4\mathcal{O}_{K_4}$, identity ray class $\mathfrak{A}$, $\mathfrak{b} = \mathcal{O}_{K_4}$, $\alpha = 4$, characteristic

$$\mathbf{r} = \begin{pmatrix} 0 \\ 1/4 \end{pmatrix}, \quad B_4 = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}^3 = \begin{pmatrix} 21 & -8 \\ 8 & -3 \end{pmatrix}.$$

Here $4(\beta_4\mathbb{Z} + \mathbb{Z}) = \mathfrak{bm}$, $4(r_2\beta_4 - r_1)\mathcal{O}_{K_4} = \beta_4\mathcal{O}_{K_4} = \mathcal{O}_{K_4} \in \mathfrak{A}$, and $r_2\beta'_4 - r_1 = \beta'_4/4 > 0$. Moreover $B_4\mathbf{r} - \mathbf{r} = (-2, -1)^\top$; B_4 is the positive generator of $\Gamma_{\mathbf{r}}$ with expanding eigenvalue β_4^3 . The fiber of

$$\mathrm{Cl}_{(4)\infty_1\infty_2}(K_4) \longrightarrow \mathrm{Cl}_{(4)\infty_2}(K_4)$$

has order two, so Kopp's exponent is $n = 2/2 = 1$.

Kopp defines a sign class $\mathfrak{R}$ in [1, Definition 6.3]. Here it is represented by $(-1 \bmod 4, + \text{ at } \infty_2)$. The residues of $\phi^n = a_n + b_n\phi$ for $0 \leq n < 6$ are

$$(a_n, b_n) = (1, 0), (0, 1), (1, 1), (1, 2), (2, 3), (3, 1),$$

and $\phi^6 \equiv 1 \pmod{4}$. Hence every unit congruent to -1 modulo 4 is $-\phi^{6k}$, which is negative at ∞_2 . Thus $\mathfrak{R} \neq \mathfrak{A}$; by (D4.6) it is the nonidentity class, and the differenced partial zeta is

$$Z_{(4)\infty_2}(s, \mathfrak{A}) = \zeta(s, \mathfrak{A}) - \zeta(s, \mathfrak{R}\mathfrak{A}) = L(s, \chi). \quad (\text{D4.8})$$

The Rademacher invariant of B_4 is zero: $s(21, 8) = -1/16$ gives $18/8 - 3 - 12(-1/16) = 0$. Thus $\psi^2(B_4) = 1$, while Kopp's theta character gives

$$\chi_{\mathbf{r}}(B_4) = \exp\left(2\pi i \frac{1}{2} \left(-\frac{28}{4} + \frac{152}{16}\right)\right) = i.$$

Consequently

$$(\psi^{-2}\chi_{\mathbf{r}}^{-1})(B_4) = -i = \Phi_{(0,1)}^2, \quad \Phi_{(0,1)} = e^{3\pi i/4}. \quad (\text{D4.9})$$

For completeness, put $C = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}$. The characteristic orbit in the three cocycle factors is

$$(0, 1) \mapsto (3, 0) \mapsto (1, 3) \mapsto (0, 1),$$

with unreduced steps

$$C(0, 1) = (-1, 0) = (3, 0) + (-4, 0), \quad C(3, 0) = (9, 3) = (1, 3) + (8, 0), \quad C(1, 3) = (0, 1).$$

Substitution in (D4.DS) gives exponent $4(0 + 1) + 0 + \min(4, 1) = 5$ and the three arguments $1 + \beta_4/4$, $1/4$, and $1 + (3\beta_4 - 1)/4$. Hence

$$\nu_{(0,1)} = \Phi_{(0,1)} \text{shin}_{B_4}^{\mathbf{r}}(\beta_4) = -x. \quad (\text{D4.10})$$

Squaring (D4.10) and applying Kopp's formula line by line now gives

$$\begin{aligned} x^2 &= \Phi_{(0,1)}^2 \text{shin}_{B_4}^{\mathbf{r}}(\beta_4)^2 \\ &= (\psi^{-2}\chi_{\mathbf{r}}^{-1})(B_4) \text{shin}_{B_4}^{\mathbf{r}}(\beta_4)^2 \\ &= \exp Z'_{(4)\infty_2}(0, \mathfrak{A}) = \exp L'(0, \chi) = u. \end{aligned} \quad (\text{D4.11})$$

Since $u + u^{-1} = 1 + \sqrt{5}$ and $x > 0$,

$$x + x^{-1} = \sqrt{3 + \sqrt{5}}.$$

Thus (D4.5) makes every minor vanish; (D4.4) then gives

$$\text{rank } K^{(4)} = 1, \quad (K^{(4)})^2 = K^{(4)}. \quad (\text{D4.12})$$

Lemma 3 (Dimension-four coefficient bridge). *For the canonical tuple $t_4 = (4, 1, \langle 1, -3, 1 \rangle)$, the equality $(K^{(4)})^2 = K^{(4)}$ is equivalent to the $\lambda = 1$ instance of AFK equation (1.49).*

Proof. For $\mathbf{p} \neq \mathbf{0}$, let $\nu_{\mathbf{p}}$ denote the analytic value from Lemma 2. Reciprocity gives

$$\nu_{\mathbf{p}}\nu_{-\mathbf{p}} = 1. \quad (\text{D4.13})$$

The actual ghost reconstruction is

$$K^{(4)} = \frac{1}{4}I + \frac{1}{4\sqrt{5}} \sum_{\mathbf{p} \neq \mathbf{0}} \nu_{\mathbf{p}} D_{\mathbf{p}}; \quad (\text{D4.14})$$

the artificial table entry $(T_4)_{0,0} = \sqrt{5}$ packages the first term and is not used as an analytic cocycle value.

Expand (D4.14) using $D_{\mathbf{p}}D_{\mathbf{q}} = \tau_4^{\langle \mathbf{p}, \mathbf{q} \rangle} D_{\mathbf{p}+\mathbf{q}}$. After multiplication by 80, the coefficient of $D_{\mathbf{p}}$ for $\mathbf{p} \neq \mathbf{0}$ is

$$\sum_{\substack{\mathbf{q} \bmod 4 \\ \mathbf{q} \neq \mathbf{0}, \mathbf{p}}} \tau_4^{\langle \mathbf{p}, \mathbf{q} \rangle} \nu_{\mathbf{p}-\mathbf{q}} \nu_{\mathbf{q}} - 2\sqrt{5} \nu_{\mathbf{p}}. \quad (\text{D4.15})$$

By (D4.13), the product is $\nu_{\mathbf{q}}/\nu_{\mathbf{q}-\mathbf{p}}$.

AFK's $\nu_{\mathbf{0}}^{\text{aux}}$ is a separate auxiliary cocycle value. Their zero-characteristic lemma gives

$$\nu_{\mathbf{0}}^{\text{aux}} + (\nu_{\mathbf{0}}^{\text{aux}})^{-1} = -(4-2)\sqrt{5} = -2\sqrt{5}. \quad (\text{D4.16})$$

Thus adjoining the omitted endpoints $\mathbf{q} = \mathbf{0}, \mathbf{p}$ to (D4.15) produces exactly the full endpoint-corrected sum

$$\sum_{\mathbf{q} \in \mathcal{I}_{\mathbf{p}}} \tau_4^{\langle \mathbf{p}, \mathbf{q} \rangle} \frac{\nu_{\mathbf{q}}}{\nu_{\mathbf{q}-\mathbf{p}}}, \quad (\text{D4.17})$$

with the endpoint ratios interpreted through $\nu_{\mathbf{0}}^{\text{aux}}$.

For nonzero characteristics write

$$\nu_{\mathbf{q}} = \Phi_{\mathbf{q}} \text{shin}_{A_{t_4}}^{\mathbf{q}/4}(\beta_4), \quad A_{t_4} = B_4.$$

The two phase identities in AFK Section 5.4 give

$$\begin{aligned} \frac{\Phi_{\mathbf{q}}}{\Phi_{\mathbf{q}-\mathbf{p}}} &= (-1)^{1+s_4(\mathbf{p})} \tau_4^{Q_4(\mathbf{p})} \tau_4^{(4-3)\langle \mathbf{p}, \mathbf{q} \rangle} \omega_4^{\langle \mathbf{p}, L_{z, t_4} \mathbf{q} \rangle}, \\ \frac{\text{shin}_{A_{t_4}}^{\mathbf{q}/4}(\beta_4)}{\text{shin}_{A_{t_4}}^{(\mathbf{q}-\mathbf{p})/4}(\beta_4)} &= \text{shin}_{A_{t_4}}^{\mathbf{q}/4}(\beta_4) \text{shin}_{A_{t_4}^{-1}}^{(\mathbf{q}-\mathbf{p})/4}(\beta_4). \end{aligned}$$

For the identity twist $f_{t_4}(1) = 1$. Its Weyl chirp contributes the additional $\tau_4^{\langle \mathbf{p}, \mathbf{q} \rangle}$, and hence

$$\tau_4^{(1+4-3)\langle \mathbf{p}, \mathbf{q} \rangle} = \omega_4^{\langle \mathbf{p}, \mathbf{q} \rangle}.$$

Up to a nonzero factor independent of $\mathbf{q}$, the coefficient equation is therefore

$$\sum_{\mathbf{q} \in \mathcal{I}_{\mathbf{p}}} \omega_4^{\langle \mathbf{p}, (I+L_{z, t_4}) \mathbf{q} \rangle} \text{shin}_{A_{t_4}}^{\mathbf{q}/4}(\beta_4) \text{shin}_{A_{t_4}^{-1}}^{(\mathbf{q}-\mathbf{p})/4}(\beta_4) = 16\delta_{\mathbf{p}, \mathbf{0}}^{(4)}. \quad (\text{D4.TCC})$$

This is exactly AFK equation (1.49) with $r = 1$ and $\lambda = 1$. For $\mathbf{p} = \mathbf{0}$, the cocycle inverse identity makes the sum 16 automatically. Every step is reversible, including the endpoint correction (D4.16). $\square$

By (D4.4) and (D4.12), $K^{(4)}$ is a trace-one rank-one idempotent, so Lemma 3 proves that 1 is a shift. The twist-shift congruence for the identity twist is

$$2\lambda + 7 \equiv 1 \pmod{8},$$

hence $\lambda \equiv 1 \pmod{4}$.

Complex conjugation replaces the twist G by $J_0 G$, where $J_0 = \text{diag}(1, -1)$. It preserves idempotence and sends

$$\lambda \longmapsto -(\lambda + d_j - 1) = 1 - \lambda \pmod{4}.$$

It therefore supplies the $\lambda = 0$ equation. Equivalently, $\det(J_0 G) = -1$ gives $-(2\lambda + 7) \equiv 1 \pmod{8}$ and hence $\lambda \equiv 0 \pmod{4}$.

Finally, dimension-four admissibility forces $r = j = m = 1$, $K_4 = \mathbb{Q}(\sqrt{5})$, and conductor one, so every admissible form has discriminant five. Class number one and the norm- -1 unit ϕ give narrow class number one; all such forms lie in one proper equivalence class. The AFK Section 7.1 transformation with $\mathbf{q} \mapsto \ell M \mathbf{q}$ gives $\mathcal{Z}_t \subseteq \mathcal{Z}_{tM}$, and applying it to M^{-1} gives the reverse inclusion. Thus both shifts hold for every dimension-four admissible tuple and their shift sets are equal.

3 Dimension five: canonical data and conventions

Put

$$K = \mathbb{Q}(\sqrt{3}), \quad \beta = 2 + \sqrt{3}, \quad Q = \langle 1, -4, 1 \rangle, \quad L = \begin{pmatrix} 4 & -1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$B = L^3 = \begin{pmatrix} 56 & -15 \\ 15 & -4 \end{pmatrix}, \quad B \begin{pmatrix} 0 \\ 1/5 \end{pmatrix} - \begin{pmatrix} 0 \\ 1/5 \end{pmatrix} = \begin{pmatrix} -3 \\ -1 \end{pmatrix}.$$

Label the real embeddings by $\infty_1(\sqrt{3}) = +\sqrt{3}$ and $\infty_2(\sqrt{3}) = -\sqrt{3}$. The modulus throughout is

$$\mathfrak{m} = (5)\infty_2.$$

We continue to use the reciprocal double-sine convention (1). For $0 \leq c \leq 5$ and $0 \leq b < 5$, write

$$\mathcal{D}_{c,b} := S_2 \left(\frac{c + b\beta}{5} \middle| \beta, 1 \right).$$

For $(p, q) \neq (0, 0)$, let $r \equiv -p - q \pmod{5}$. The principal-form specialization of the AFK three-factor formula is

$$\tilde{\nu}_{p,q} = (-1)^{5(p+q)+pq+\min(5,p+q)} \mathcal{D}_{5-p,q} \mathcal{D}_{5-r,p} \mathcal{D}_{5-q,r}. \quad (2)$$

Formula (2) is used only for nonzero characteristics. Later we set the table entry $T_{0,0} := \sqrt{6}$ solely so that the identity Weyl coefficient is $T_{0,0}/\sqrt{6} = 1$. This reconstruction convention is not AFK's auxiliary zero-characteristic cocycle value.

Lemma 4 (Derivation and convention check for (2)). *Formula (2) is the specialization of AFK Definition 1.32 to $t = (5, 1, \langle 1, -4, 1 \rangle)$, with no unrecorded reciprocal or sign.*

Proof. AFK Definition 1.32 first forms $\tilde{\nu}_{\mathbf{p}} = \Phi_{\mathbf{p}} \text{shin}_B^{\mathbf{p}/5}(\beta)$. Insert the defining modular-to-Jacobi quotient [2, equation (1.26)], and then use the Jacobi cocycle decomposition [2, equations (8.2)–(8.5)] and the S -kernel formula [2, equation (8.6)]. In the notation of AFK Section 8.2,

$$L = T^4 S, \quad B = L^3, \quad L\beta = \beta, \quad j_{L^k}(\beta) = \beta^k.$$

Thus the word $B = (T^4 S)^3$ contributes exactly three S -kernels. Modulo five the characteristic orbit is

$$(p, q) \mapsto (r, p) \mapsto (q, r) \mapsto (p, q), \quad r \equiv -p - q \pmod{5},$$

and the three unreduced linear arguments are

$$\frac{q\beta - p}{5}, \quad \frac{p\beta - r}{5}, \quad \frac{r\beta - q}{5}.$$

Reducing their first coordinates to the representatives $5 - p, 5 - r, 5 - q$ gives respectively $\mathcal{D}_{5-p,q}$, $\mathcal{D}_{5-r,p}$, and $\mathcal{D}_{5-q,r}$. AFK's S -kernel is its displayed double sine in the denominator (AFK Section 8.2, equation for σ_S). That double sine is Kopp's $\text{Sin}_{2,\text{Kopp}}$; hence each denominator is precisely one factor of the reciprocal convention (1).

For completeness, apply the integral-translation law [2, equation (8.9)] to the three representative changes and multiply by the phase in AFK Definition 1.30. Direct substitution gives

$$\Phi_{p,q}(\text{three translation multipliers}) = (-1)^{E(p,q)}, \quad E(p,q) = 5(p+q) + pq + \min(5, p+q).$$

This is a finite parity calculation:

$$E(p,q) \equiv \begin{cases} pq & (p+q \leq 5), \\ p+q+pq+1 & (p+q \geq 5) \end{cases} \pmod{2},$$

where the two expressions agree at $p+q=5$. In the order $p=0, \dots, 4$ (columns $q=0, \dots, 4$), the resulting signs are

$$\begin{pmatrix} * & + & + & + & + \\ + & - & + & - & + \\ + & + & + & + & - \\ + & - & + & + & + \\ + & + & - & + & - \end{pmatrix};$$

the asterisk is the excluded zero characteristic. This accounts for all twenty-four cases and proves (2) directly from the cited definitions. $\square$

4 Ray labels and all characteristics

Lemma 5 (Certified ray arithmetic). *The class and ray groups are*

$$\text{Cl}(K) = 1, \quad \text{Cl}_{(5)}(K) \cong C_4, \quad \text{Cl}_{\mathfrak{m}}(K) \cong C_8, \quad \text{Cl}_{(5)\infty_1\infty_2}(K) \cong C_8 \times C_2. \quad (3)$$

Kopp's residue/sign class $\mathfrak{R}$ is the element $4 \in C_8$. The fiber of $\text{Cl}_{(5)\infty_1\infty_2}(K) \rightarrow \text{Cl}_{\mathfrak{m}}(K)$ has order two, so the exponent in Kopp's theorem is $n=2/2=1$. The computation is unconditional.

Proof. PARI/GP 2.15.4 gives the structures in (3), and `bnfcertify=1`. The unit group is $\{\pm\beta^n : n \in \mathbb{Z}\}$ and $\beta^3 \equiv 1 \pmod{5}$. Hence units congruent to -1 modulo 5 are $-\beta^{3k}$. Their value under ∞_2 is $-(2 - \sqrt{3})^{3k} < 0$, so no global unit realizes the pair $(-1 \pmod{5}, + \text{ at } \infty_2)$ defining Kopp's $\mathfrak{R}$. It is therefore the nontrivial sign class, namely the unique element of order two in C_8 . $\square$

Proposition 6 (No quadratic-character shortcut). *The Kopp partial-zeta difference has zero coefficient at the unique order-two character of $\text{Cl}_{\mathfrak{m}}(K)$. It is supported on four characters, all of order eight.*

Proof. Write $\text{Cl}_{\mathfrak{m}}(K) = \langle g \rangle \cong C_8$ and $\mathfrak{R} = g^4$. For $\chi_k(g) = \exp(2\pi i k/8)$, Fourier inversion gives

$$\zeta_{\mathfrak{m}}(s, \mathfrak{A}) - \zeta_{\mathfrak{m}}(s, \mathfrak{R}\mathfrak{A}) = \frac{1}{8} \sum_{k=0}^7 \overline{\chi_k(\mathfrak{A})} (1 - \overline{\chi_k(\mathfrak{R})}) L(s, \chi_k). \quad (\text{CF})$$

Since $\chi_k(\mathfrak{R}) = (-1)^k$, the coefficient is zero for even k and two for odd k . The support is therefore $k=1, 3, 5, 7$, whose characters all have order eight. The unique quadratic character is χ_4 ; it is even on $\mathfrak{R}$ and its coefficient in (CF) is zero. Hence the difference cannot factor through the quadratic quotient. This is the precise reason the dimension-four quadratic L -function reduction does not recur. Equivalently, $\mathfrak{R} = g^4$ lies in the kernel $\langle g^2 \rangle$ of the quotient $C_8 \rightarrow C_2$, so $\mathfrak{A}$ and $\mathfrak{R}\mathfrak{A}$ have the same image in the unique quadratic quotient and their difference projects to zero. $\square$

For each $(p, q) \neq (0, 0)$ start with p and repeatedly subtract five; let $\tilde{p}$ be the first lift for which

$$q(2 - \sqrt{3}) - \tilde{p} > 0. \quad (4)$$

Let $\mathfrak{q}$ be the prime above 3, and use it as the generator of C_8 . PARI's raw cyclic basis g gives the discrete logarithms of the positive principal ideals $(q\beta - \tilde{p})$ in the audit transcript, with $\mathfrak{q} = g^5$. Multiplication by 5, which is its own inverse modulo 8, converts those raw logs to the normalized $\mathfrak{q}$ -coordinates below:

$p \backslash q$	0	1	2	3	4
0	—	0	6	2	4
1	4	7	5	7	0
2	2	5	5	6	3
3	6	7	2	1	1
4	0	4	3	1	3

(5)

The corresponding positive lifts are 0 in the first row, -4 in row one except at $(1, 4)$ where it is 1, and $-3, -2, -1$ in rows two, three, and four.

Lemma 7 (All multiplier comparisons). *For every nonzero $(p, q) \bmod 5$, Kopp's modular multiplier equals the square of the AFK phase at that characteristic.*

Proof. Write $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\mathbf{r} = (\tilde{p}/5, q/5)^\top$, with $\tilde{p}$ chosen by (4). Substitution in Kopp's defining theta-character gives the exponent

$$\frac{1}{2} \{ (c - d + 1)r_1 + (-a + b + 1)r_2 - cdr_1^2 + 2(a - 1)dr_1r_2 - (a - 2)br_2^2 \}. \quad (6)$$

For $B = \begin{pmatrix} 56 & -15 \\ 15 & -4 \end{pmatrix}$, direct reduction of (6) modulo one gives

$$\frac{p^2 - 4pq + q^2}{5} \pmod{1} \quad (7)$$

in all twenty-four cases. Also $s(-4, 15) = -19/90$, so the Rademacher invariant is three and $\psi^2(B) = i$. Therefore the Kopp multiplier has exponent

$$-\frac{1}{4} - \frac{p^2 - 4pq + q^2}{5} \pmod{1}, \quad (8)$$

which is precisely the exponent of the square of the AFK phase. The exact twenty-four-row substitution, including every positive lift, is part of the bridge certificate. $\square$

For a normalized ray log k , name the positive square root of its ray value by

k	0	1	2	3	4	5	6	7
positive root	x	w	y^{-1}	z	x^{-1}	w^{-1}	y	z^{-1}

The class log in (5) selects the unsigned entry and the explicit sign in (2) selects its sign. Thus the complete reconstruction table is

$$T_5 = \begin{pmatrix} \sqrt{6} & x & y & y^{-1} & x^{-1} \\ x^{-1} & -z^{-1} & w^{-1} & -z^{-1} & x \\ y^{-1} & w^{-1} & w^{-1} & y & -z \\ y & -z^{-1} & y^{-1} & w & w \\ x & x^{-1} & -z & w & -z \end{pmatrix}. \quad (9)$$

Here x, y, z, w denote the four positive representatives of the ray-class pairs

$$(0, 4), \quad (6, 2), \quad (3, 7), \quad (1, 5), \quad (10)$$

respectively. In particular, the table is generated from the ray logs and the sign exponent; it is not stored as a list of expected answers in the bridge generator.

5 The analytic ray packet

Set

$$\begin{aligned}
P(U) = & U^8 - (8 + 5\sqrt{3})U^7 + (53 + 30\sqrt{3})U^6 \\
& - (156 + 90\sqrt{3})U^5 + (225 + 130\sqrt{3})U^4 \\
& - (156 + 90\sqrt{3})U^3 + (53 + 30\sqrt{3})U^2 \\
& - (8 + 5\sqrt{3})U + 1.
\end{aligned} \tag{11}$$

It is the reciprocal lift $P(U) = U^4 F(U + U^{-1})$ of

$$\begin{aligned}
F(V) = & V^4 - (8 + 5\sqrt{3})V^3 + (49 + 30\sqrt{3})V^2 \\
& - (132 + 75\sqrt{3})V + (121 + 70\sqrt{3}).
\end{aligned} \tag{12}$$

Kopp previously recognized the same polynomial numerically as the Stark-unit polynomial for $\mathbb{Q}(\sqrt{3})$ and modulus five; see [4, §7, (7.21)] and [5, Example 1.17, (3.12)]. PARI's `bnrstark` reproduces (12), but that recognition is not used as a proof below.

The square roots of the reciprocal packet have absolute polynomial

$$\begin{aligned}
Q(X) = & X^{32} - 16X^{30} + 95X^{28} - 260X^{26} + 355X^{24} - 348X^{22} \\
& + 388X^{20} - 300X^{18} + 195X^{16} - 300X^{14} + 388X^{12} \\
& - 348X^{10} + 355X^8 - 260X^6 + 95X^4 - 16X^2 + 1.
\end{aligned} \tag{13}$$

Define $w^{-1}, z^{-1}, x^{-1}, y^{-1}, y, x, z, w$ to be the unique positive roots of (13) in, respectively,

$$\begin{aligned}
& (0.424835, 0.424836), (0.481476, 0.481477), (0.506963, 0.506964), (0.782390, 0.782391), \\
& (1.278133, 1.278134), (1.972526, 1.972527), (2.076946, 2.076947), (2.353849, 2.353850).
\end{aligned} \tag{14}$$

Sturm's theorem certifies exactly one root in every interval.

5.1 Shintani's theorem with an explicit exponent

For a ray class $A \in \text{Cl}_{\mathfrak{m}}(K)$ put

$$X_A := \exp Z'_{\mathfrak{m}}(0, A).$$

Lemma 8 (Specialized Shintani algebraicity). *For every $A \in \text{Cl}_{\mathfrak{m}}(K)$,*

$$X_A^{5760} \in \mathcal{O}_H^{\times},$$

where H is the ray field of modulus $\mathfrak{m}$. These powers have the Artin action indexed by the ray classes. At every embedding of H inducing ∞_2 on K , their absolute value is one.

Proof. Work first in the full narrow ray group

$$C := \text{Cl}_{(5)\infty_1\infty_2}(K) \cong C_8 \times C_2.$$

In Shintani's notation, let μ be the class represented by $11 - 5\beta$, which is congruent to 1 modulo 5 and has signs $(-, +)$ at (∞_1, ∞_2) . Let ν be the class represented by 4, which is congruent to -1 modulo 5 and is totally positive. The exact ray logs are

$$\mu = [4, 1] \in C, \quad \nu = [4, 0] \in C.$$

Forgetting ∞_1 kills μ and sends ν to the unique order-two element $4 \in C_8$. Hence

$$G = \langle \mu \rangle, \quad \nu \notin G, \quad C/G \simeq \text{Cl}_{\mathfrak{m}}(K).$$

Every unit is $\pm\beta^n$, and β has order three modulo 5. Thus no totally positive unit is -1 modulo 5, which is Shintani's condition (0-3). There is no norm- -1 unit, since $a^2 - 3b^2 = -1$ is impossible modulo 3; this implies condition (0-6).

The field H has signature $(8, 4)$ and a unique octic subfield

$$M \simeq \mathbb{Q}(\zeta_{60})^+.$$

Thus H/M is quadratic, M is the maximal absolutely abelian subfield of H , and exactly one real place of K splits in H . These are Shintani's hypothesis (0-9). His Theorem 2 [6] therefore applies to $X_{(5)}(c, G)$.

This invariant is exactly the one-place zeta difference used here. Indeed, if c maps to A , then the inverse images of A and RA in C are respectively

$$\{c, c\mu\}, \quad \{c\nu, c\mu\nu\}.$$

Shintani's Theorem 1 consequently gives

$$\begin{aligned} \log X_{(5)}(c, G) &= \zeta'_{\text{full}}(0, c) - \zeta'_{\text{full}}(0, c\nu) \\ &\quad + \zeta'_{\text{full}}(0, c\mu) - \zeta'_{\text{full}}(0, c\mu\nu) \\ &= \zeta'_{\mathfrak{m}}(0, A) - \zeta'_{\mathfrak{m}}(0, RA) = Z'_{\mathfrak{m}}(0, A). \end{aligned}$$

It remains to make Shintani's existential power explicit. The normal closure of H is the full two-infinite-place ray field. It is also the degree-sixteen ray field over $\mathbb{Q}(\sqrt{-5})$ with conductor

$$\mathfrak{c} = \begin{pmatrix} 15 & 0 \\ 0 & 3 \end{pmatrix}$$

in PARI's integral basis. The quadratic-subfield enumeration contains a unique copy of $\mathbb{Q}(\sqrt{-5})$, so this identification preserves the imaginary quadratic field occurring in Shintani's proof.

Shintani's Proposition 4 expresses the real invariant as a product of ratios of imaginary-quadratic W -invariants with integral multiplicities. For a nontrivial divisor $\mathfrak{d} \mid \mathfrak{c}$, clearing the absolute-value square root and the distribution exponent requires a multiple of $12f_{\mathfrak{d}}n(S)$. For the trivial divisor the required exponent is $12h_k n(S)$. The exact exponents for the eight divisors are

$$384, 288, 288, 144, 240, 360, 360, 180,$$

whose least common multiple is 5760. Since (5) is prime and its only proper divisor fails (0-3), Shintani's $Y_{(5)}$ equals $X_{(5)}$ and no induction exponent is added. Proposition 5 supplies the unit, Artin, and nonsplit-modulus-one assertions. $\square$

5.2 The labeled candidate field

The absolute polynomial of a root of (11) is

$$\begin{aligned} p(T) &= T^{16} - 16T^{15} + 95T^{14} - 260T^{13} + 355T^{12} - 348T^{11} \\ &\quad + 388T^{10} - 300T^9 + 195T^8 - 300T^7 + 388T^6 - 348T^5 \\ &\quad + 355T^4 - 260T^3 + 95T^2 - 16T + 1. \end{aligned}$$

Lemma 9 (Exact field and Frobenius labels). *The field defined by (11) is the labeled ray field H over K . Let U be the root of p in*

$$(3.890, 3.891).$$

Then U is a norm-one unit. Arithmetic Frobenius at the prime $\mathfrak{q}$ above 3 acts transitively on its eight real conjugates, which lie successively in

$$(3.890, 3.891), (5.540, 5.541), (0.612, 0.613), (4.313, 4.314), \\ (0.257, 0.258), (0.180, 0.181), (1.633, 1.634), (0.231, 0.232).$$

At the embeddings above ∞_2 , all conjugates of U have absolute value one.

Proof. Applying `rnfequation` to both the candidate and the ray field, `nfisism` returns eight maps. Exact substitution shows that all eight maps fix the labeled root of $y^2 - 4y + 1$; hence the isomorphism is over K , not merely over $\mathbb{Q}$. Moreover `bnfcertify=1`, `bnfisunit` gives an exact unit decomposition, and the norm of U is one.

The prime $\mathfrak{q}$ generates the ray group. Modulo $\mathfrak{q}$ the relative polynomial is

$$T^8 + T^7 + 2T^6 + 2T^2 + T + 1 \in \mathbb{F}_3[T],$$

which is irreducible. Arithmetic Frobenius is therefore $T \mapsto T^3$. Reduction modulo this polynomial identifies one of the exact `nfgaloisconj` automorphisms with Frobenius.

A rational interval of width 10^{-80} contains exactly one root of p in $(3.890, 3.891)$. Exact rational interval evaluation of the successive Frobenius polynomials places their images inside the eight disjoint intervals in the statement. This proves the labels Galois-equivariantly.

Finally, under the nonidentity embedding of K , substitute $T = 2(Y^2 - 1)/(Y^2 + 1)$ in the trace polynomial (12). The exact `nfpolsturm` counts are $[8, 0]$, so all four trace roots at that embedding lie in $(-2, 2)$. Each corresponding reciprocal quadratic has both roots on the unit circle. $\square$

5.3 Certified removal of the power

Theorem 10 (Unconditional ray packet). *For*

$$\mathcal{X}_k := \exp Z'_m(0, \mathfrak{q}^k), \quad 0 \leq k < 8,$$

the exact Artin-labeled packet is

$$(\mathcal{X}_0, \dots, \mathcal{X}_7) = (x^2, w^2, y^{-2}, z^2, x^{-2}, w^{-2}, y^2, z^{-2}). \quad (15)$$

In particular, $\mathcal{X}_0 = U$.

Proof. Kopp's proved Kronecker-limit theorem, with the convention and multiplier checked above, gives

$$\tilde{\nu}_{p,q}^2 = \exp Z'_m(0, \mathfrak{A}_{p,q}); \quad (16)$$

here $\mathfrak{A}_{p,q}$ is the normalized ray class in (5). The four positive generators at characteristics

$$(0, 1), \quad (0, 2), \quad (2, 4), \quad (3, 3)$$

have $\mathfrak{q}$ -coordinates 0, 6, 3, 1, respectively.

We evaluate the regularized integral for the reciprocal double sine (1) with Arb balls through `python-flint` [9]. Every Simpson panel uses an Arb enclosure of the fourth derivative. All endpoints, widths, tolerances, and acceptance comparisons are exact rationals or outward-rounded balls. Near zero, the positive series of $H_c(t) = \sinh(ct/2)/(ct/2)$ gives the rational bound

$$|f(t) - f(0)| \leq 4t^2 \quad (0 \leq t \leq 10^{-4}).$$

For the tail, every reduced argument and its complement belongs to $(1/100, 5)$; beyond 36, each exponential denominator exceeds $3/4$, giving the remainder bound $e^{-36}/9$. The candidate roots are selected by disjoint rational isolating intervals, independently of library ordering.

For each of the four characteristics, the ball for $\log(\tilde{\nu}_{p,q}^2) - \log(U_k)$ contains zero. The largest radius is less than

$$\epsilon = 2.23 \times 10^{-9}.$$

The reciprocal partners give the remaining four real embeddings.

By Lemma 8, both $\mathcal{X}_0^{5760}$ and U^{5760} are units in the same labeled field. Set

$$\eta = \mathcal{X}_0^{5760}/U^{5760}.$$

Lemma 9 and the interval labels show that at every real embedding,

$$|\log |\sigma(\eta)|| < 5760\epsilon < 1.29 \times 10^{-5}.$$

At the four complex pairs, both numerator and denominator have absolute value one. Hence

$$h(\eta) < 1.29 \times 10^{-5}.$$

If $3 \leq d = [\mathbb{Q}(\eta) : \mathbb{Q}] \leq 16$ and η is not a root of unity, Voutier's explicit bound [7] gives

$$h(\eta) > \frac{1}{4d} \left(\frac{\log \log d}{\log d} \right)^3.$$

Its minimum for $3 \leq d \leq 16$ is $5.2279533222 \dots \times 10^{-5}$, a contradiction. A non-torsion quadratic unit has height at least $\frac{1}{2} \log((1 + \sqrt{5})/2)$, and a rational unit is ± 1 . Thus η is a root of unity. Since H has a real embedding, its only roots of unity are ± 1 ; positivity at the distinguished embedding gives $\eta = 1$. Therefore $\mathcal{X}_0^{5760} = U^{5760}$ and positivity yields $\mathcal{X}_0 = U$. The exact Artin action in Lemma 8, together with the certified Frobenius orbit in Lemma 9, proves the full packet (15). $\square$

Since every reciprocal double-sine factor in (2) is positive at the displayed arguments, equations (2), (5), and (15) now derive the signed table (9) unconditionally.

6 Exact finite cancellation

Let $\zeta_5 = e^{2\pi i/5}$ and $\tau = -e^{\pi i/5} = \zeta_5^3$. Define, without reference to software,

$$\Pi_{r,c}^{(5)} := \frac{1}{5\sqrt{6}} \sum_{\substack{0 \leq a,b < 5 \\ r \equiv c+a \pmod{5}}} T_{a,b} \tau^{ab} \zeta_5^{bc}, \quad 0 \leq r, c < 5, \quad (17)$$

where T is (9). Equivalently, $\Pi^{(5)} = \frac{1}{5} \sum_{a,b} (T_{a,b}/\sqrt{6}) D_{a,b}$ for $D_{a,b} e_c = \tau^{ab} \zeta_5^{bc} e_{c+a}$. In particular,

$$\text{Tr } \Pi^{(5)} = T_{0,0}/\sqrt{6} = 1. \quad (18)$$

Proposition 11 (Exact 100-minor certificate). *For the algebraic values selected by (13)–(14), all one hundred 2-minors of the matrix (17) vanish exactly. Hence $\text{rank } \Pi^{(5)} = 1$ and $(\Pi^{(5)})^2 = \Pi^{(5)}$.*

Proof. Over the coefficient field $\mathbb{Q}(\zeta_5, \sqrt{6})$, (13) has four degree-eight factors. Let w be the interval-defined root in (14). Exact rational interval arithmetic applied to the `nfgaloisconj` polynomials proves that `nfgaloisconj` substitutions are

	w	w^{-1}	y	y^{-1}	x	x^{-1}	z	z^{-1}
conjugate index	2	8	5	4	10	6	16	12.

(19)

with every image lying inside its corresponding interval in (14). Moreover, exact `nfisincl` identities express $\sqrt{5}$ and $\sqrt{6}$ as polynomials in w ; rational interval evaluation selects their positive signs. Among the four factors, factor four is the unique factor on which both of these subfield elements equal the positive $\sqrt{5}$ and positive $\sqrt{6}$. This certifies, rather than numerically assumes, that the interval-defined tuple is the tuple used in the factor-four calculation.

Substitution in (9) and (17), followed by exact reduction modulo the selected factor, makes all one hundred minors zero. As a factor selection check, factors one, two, and three leave respectively 70, 100, 100 nonzero minors. This is exact number-field arithmetic, not floating-point evaluation. The complete script and transcript are supplied with the manuscript.

The matrix is nonzero because (18) gives trace one. Thus its rank is exactly one. Any rank-one matrix M satisfies $M^2 = (\text{Tr } M)M$, proving idempotency. $\square$

7 Dimension-five twisted convolution

Lemma 12 (Weyl coefficient bridge). *The matrix $\Pi^{(5)}$ of (17) is the standard-basis Weyl reconstruction of the canonical AFK ghost for $t = (5, 1, \langle 1, -4, 1 \rangle)$. Moreover, $(\Pi^{(5)})^2 = \Pi^{(5)}$ is equivalent to the $\lambda = 1$ instance of AFK equation (1.49):*

$$\sum_{\mathbf{q} \in \mathcal{I}_{\mathbf{p}}} \zeta_5^{\langle \mathbf{p}, (I+L_{z,t})\mathbf{q} \rangle} \text{shin}_{A_t}^{\mathbf{q}/5}(\beta) \text{shin}_{A_t^{-1}}^{(\mathbf{q}-\mathbf{p})/5}(\beta) = 25\delta_{\mathbf{p}, \mathbf{0}}^{(5)}. \quad (20)$$

Proof. For $\mathbf{p} \neq \mathbf{0}$, put $\nu_{\mathbf{p}} := \tilde{\nu}_{\mathbf{p}}$ as given by (2). The first two unconditional parts of AFK's construction give

$$\nu_{\mathbf{p}}\nu_{-\mathbf{p}} = 1. \quad (21)$$

Theorem 10, followed by (2) and (5), identifies these twenty-four numbers with the nonzero entries of T . The actual ghost reconstruction is therefore

$$\Pi^{(5)} = \frac{1}{5}I + \frac{1}{5\sqrt{6}} \sum_{\mathbf{p} \neq \mathbf{0}} \nu_{\mathbf{p}} D_{\mathbf{p}}. \quad (22)$$

The artificial entry $T_{0,0} = \sqrt{6}$ packages the first term of (22) into the single entry formula (17); it is not an analytic zero-characteristic value.

Now expand $(\Pi^{(5)})^2 - \Pi^{(5)}$ using $D_{\mathbf{p}}D_{\mathbf{q}} = \tau^{\langle \mathbf{p}, \mathbf{q} \rangle} D_{\mathbf{p}+\mathbf{q}}$ and compare every Weyl coefficient. For $\mathbf{p} \neq \mathbf{0}$, multiplication by 150 makes its coefficient

$$\sum_{\substack{\mathbf{q} \bmod 5 \\ \mathbf{q} \neq \mathbf{0}, \mathbf{p}}} \tau^{\langle \mathbf{p}, \mathbf{q} \rangle} \nu_{\mathbf{p}-\mathbf{q}} \nu_{\mathbf{q}} - 3\sqrt{6} \nu_{\mathbf{p}}. \quad (23)$$

Using (21), the product in (23) is $\nu_{\mathbf{q}}/\nu_{\mathbf{q}-\mathbf{p}}$.

AFK's $\nu_{\mathbf{0}}^{\text{aux}}$ is an auxiliary cocycle value, not the identity coefficient in (22). Their zero-characteristic lemma specializes to

$$\nu_{\mathbf{0}}^{\text{aux}} + (\nu_{\mathbf{0}}^{\text{aux}})^{-1} = -(5-2)\sqrt{6} = -3\sqrt{6}. \quad (24)$$

After adjoining the two omitted endpoints $\mathbf{q} = \mathbf{0}, \mathbf{p}$, their combined contribution is

$$\nu_{\mathbf{p}}(\nu_{\mathbf{0}}^{\text{aux}} + (\nu_{\mathbf{0}}^{\text{aux}})^{-1}) = -3\sqrt{6} \nu_{\mathbf{p}}.$$

Thus (23) is exactly the full endpoint-corrected sum

$$\sum_{\mathbf{q} \in \mathcal{I}_{\mathbf{p}}} \tau^{\langle \mathbf{p}, \mathbf{q} \rangle} \frac{\nu_{\mathbf{q}}}{\nu_{\mathbf{q}-\mathbf{p}}}, \quad (25)$$

with the two endpoint ratios interpreted using ν_0^{aux} .

Write $\Phi_{\mathbf{q}}$ for the AFK phase. Their phase-ratio calculation specializes, with $r = f_j/f = 1$ and $d_j = d = 5$, to

$$\frac{\Phi_{\mathbf{q}}}{\Phi_{\mathbf{q}-\mathbf{p}}} = (-1)^{1+s_5(\mathbf{p})} \tau^{Q(\mathbf{p})} \tau^{\langle \mathbf{p}, \mathbf{q} \rangle} \zeta_5^{\langle \mathbf{p}, L_{z,t} \mathbf{q} \rangle}. \quad (26)$$

The first two factors are nonzero and independent of $\mathbf{q}$. Multiplying the remaining τ factor by the displacement chirp in (25), and using $\tau^2 = \zeta_5$, gives exactly

$$\zeta_5^{\langle \mathbf{p}, (I+L_{z,t}) \mathbf{q} \rangle}.$$

The cocycle inverse identity

$$\frac{\text{shin}_{A_t}^{\mathbf{q}/5}(\beta)}{\text{shin}_{A_t}^{(\mathbf{q}-\mathbf{p})/5}(\beta)} = \text{shin}_{A_t}^{\mathbf{q}/5}(\beta) \text{shin}_{A_t^{-1}}^{(\mathbf{q}-\mathbf{p})/5}(\beta) \quad (27)$$

gives the two factors printed in (20). Here $A_t = B$, $L_{z,t} = L$, and $f_t(1) = 2 + 9 \equiv 1 \pmod{5}$, so the identity twist produces exactly the chirp in (25). The remaining factor depends on $\mathbf{p}$ but is nonzero and independent of $\mathbf{q}$. Hence the vanishing of every nonzero Weyl coefficient is equivalent to (20) for $\mathbf{p} \neq \mathbf{0}$. For $\mathbf{p} = \mathbf{0}$, the cocycle inverse identity makes the sum 25 automatically. This is the specialization of AFK equations (5.78)–(5.98), including rather than suppressing their endpoint correction. $\square$

Theorem 13 (Complete dimension-five TCC). *For every dimension-five admissible tuple t ,*

$$0, 1 \in \mathcal{Z}_t.$$

Furthermore $\mathcal{Z}_{t'} = \mathcal{Z}_t$ for any two dimension-five admissible tuples t, t' .

Proof. Proposition 11 and Lemma 12 prove the $\lambda = 1$ identity for the canonical tuple. The twist condition is

$$\det(G)f_t(\lambda) \equiv 1 \pmod{5}, \quad f_t(\lambda) = 2\lambda + 9.$$

Thus $G = I$ gives $2\lambda + 9 \equiv 1 \pmod{5}$, or $\lambda \equiv 1 \pmod{5}$.

Complex conjugation preserves idempotency and sends λ to $\bar{\lambda} = -(\lambda + d - 1) = 1 - \lambda \pmod{5}$. It therefore sends the verified shift 1 to shift 0. Equivalently the conjugate twist has determinant -1 , and $-(2\lambda + 9) \equiv 1 \pmod{5}$ gives $\lambda \equiv 0 \pmod{5}$.

Finally, dimension-five admissibility forces $r = 1$, $K = \mathbb{Q}(\sqrt{3})$, and $j = m = 1$. Since the relevant conductor is one, every admissible form has discriminant twelve. The ordinary class number is one, so primitive indefinite forms of discriminant twelve constitute one $\text{GL}_2(\mathbb{Z})$ -equivalence class. (The narrow class number is not needed: determinant- -1 transformations are allowed.) Thus every admissible form is integrally equivalent to the canonical one in the sense required here.

For clarity, the transport does not invoke AFK's conditional transformation theorem. If t_M is obtained from t by $M \in \text{GL}_2(\mathbb{Z})$, then unconditionally

$$A_{t_M} = M^{-1}A_t M, \quad L_{z,t_M} = M^{-1}L_{z,t} M, \quad \beta_{t_M} = M^{-1}\beta_t.$$

Kopp's covariance formula sends each cocycle factor at $(t_M, \mathbf{q})$ to the corresponding factor at $(t, \ell M \mathbf{q})$, with $\ell = \text{sgn}(j_{M^{-1}}(\beta_t))$; when $\det M = -1$ both factors are complex-conjugated, which does not change the real TCC sum. Reindexing

$$\mathbf{q}' = \ell M \mathbf{q}, \quad \mathcal{I}'_{\ell M \mathbf{p}} = \ell M \mathcal{I}_{\mathbf{p}},$$

and using symplectic covariance transforms the complete left side of (20) for $(t_M, \mathbf{p})$ into that for $(t, \ell M \mathbf{p})$. Hence $\mathcal{Z}_t \subseteq \mathcal{Z}_{t_M}$. Applying the same calculation to M^{-1} gives the reverse inclusion and therefore equality. $\square$

Proof of Theorem 1. The dimension-four assertion is proved in Section 2; the dimension-five assertion is Theorem 13. $\square$

8 Composite dimensions six and eight

The preceding argument succeeds because Shintani's 1978 theorem supplies powered algebraicity for the entire dimension-five packet. The same finite reconstruction can be completed in dimensions six and eight, but none of the special-value theorems used in this paper supplies the needed oriented algebraicity for every primitive characteristic. This section records computationally exact finite results, the unconditional analytic strata covered by the cited theorems, and the remaining research targets. It contains no dimension-six or dimension-eight TCC theorem.

8.1 Dimension six: one primitive order-six value

Put

$$K_6 = \mathbb{Q}(\sqrt{21}), \quad \beta_6 = \frac{5 + \sqrt{21}}{2}, \quad Q_6 = \langle 1, -5, 1 \rangle.$$

The relevant one-place ray group is

$$\mathrm{Cl}_{(6)\infty_2}(K_6) \cong C_6. \quad (28)$$

Its order-two character gives the lower-conductor stratum. If

$$Y = \frac{\beta_6 - 2 + \sqrt{\beta_6 - 1}}{2} > 1,$$

then the modulus-three audit `scripts/dimension_six_lower_stratum.gp`, combining the analytic class-number formula with Kopp's Theorem 1.1 [1], proves unconditionally that the overlap $y = \nu_{(0,2)}$ satisfies

$$y^2 = Y, \quad y^2 + y^{-2} = \beta_6 - 2. \quad (29)$$

The remaining three reciprocal pairs form the primitive order-six packet. Let

$$\begin{aligned} \mathcal{Q}_6(X) = & X^{12} + 3X^{11} - 6X^{10} - 16X^9 + 3X^8 + 27X^6 \\ & + 3X^4 - 16X^3 - 6X^2 + 3X + 1, \end{aligned} \quad (30)$$

and let x_{alg} be its unique root in

$$2.21288528901718 < x_{\mathrm{alg}} < 2.21288528901719. \quad (31)$$

The norm-37 prime $(4\beta_6 + 1)$ generates the ray group and fixes the arithmetic-Frobenius order of the other two primitive pairs. Inserting this Artin-labeled algebraic packet together with (29) in the six-dimensional Weyl reconstruction makes all

$$\binom{6}{2}^2 = 225$$

two-by-two minors vanish exactly; the reconstructed matrix has trace one. Thus the finite algebra has no remaining ambiguity.

In the double-sine convention (1), the distinguished $(0, 1)$ analytic value is

$$x_{\mathrm{an}} = \frac{S_2(\beta_6/6 \mid \beta_6, 1) S_2(1/6 \mid \beta_6, 1)}{S_2((\beta_6 + 1)/6 \mid \beta_6, 1)}. \quad (32)$$

Exact closure target. The exact finite audit shows the following conditional implication: the canonical analytic table agrees with the certified rank-one algebraic table, and hence gives both formal shifts, if

$$x_{\text{an}} = x_{\text{alg}}. \quad (33)$$

The characteristic orbits, reciprocity, and the arithmetic-Frobenius labeling propagate this one equality through the other primitive pairs. The script `scripts/dimension_six_primitive_fourier_audit.gp` records the equivalent oriented order-six Fourier identity, including its ray-group generator, character convention, Euler-factor convention, and embedding order. That analytic identity, and therefore (33), remains unproved here.

This is not a finite-computation issue. The ray field has signature $(6, 3)$; its primitive character has order six; and its cubic component is wildly ramified above 3. Shintani's index-two theorem does not apply, Roblot's hypotheses [10] do not cover this wild sextic case, and proved CM Brumer–Stark results do not apply to this mixed-signature extension. Standard double-sine shifts, reflection, duplication, and conductor lowering determine norms of the four level-six lifts but do not isolate the selected lift.

There is also no weaker local-divisibility shortcut. For

$$A_6 = \begin{pmatrix} 115 & -24 \\ 24 & -5 \end{pmatrix},$$

write $A_6 \cdot z = (115z - 24)/(24z - 5)$. Then

$$A_6 \cdot z - z = -24 \frac{z^2 - 5z + 1}{24z - 5}. \quad (34)$$

The right side has a simple zero at β_6 . Consequently, in the local holomorphic ring at β_6 , divisibility of a defect entry by $A_6 \cdot z - z$ is equivalent to its vanishing at β_6 ; proving such divisibility would already prove the missing TCC identity.

8.2 Dimension eight: an unconditional lower stratum and two quartic phases

Let

$$K_8 = \mathbb{Q}(\sqrt{5}), \quad \phi = \frac{1 + \sqrt{5}}{2}, \quad \beta_8 = 3\phi + 2 = \frac{7 + 3\sqrt{5}}{2}.$$

The canonical form has discriminant 45 and belongs to the order

$$\mathcal{O}_3 = \mathbb{Z} + 3\mathcal{O}_{K_8}.$$

The order-ray group and its maximal-order realization are

$$\text{Cl}_{(8)\infty_2}(\mathcal{O}_3) \cong \text{Cl}_{(24)\infty_2}(\mathcal{O}_{K_8}) \cong C_4 \times C_2 \times C_2. \quad (36)$$

The change-of-order isomorphism in (36) does not identify partial zeta functions by itself. Kopp's conductor-lowering formula [1] supplies the required three-factor bridge between the order-modulus-eight characteristics and maximal-order moduli 8 and 24.

The fifteen nonzero lower-conductor characteristics lie in the one-place ray-12 field. Its one-place and both-place ray groups are C_2^2 and C_2^3 , and the extension is in Shintani's index-two range. An exact divisor audit gives safe exponent 576. Arb enclosures and the height gap used above then prove unconditionally that its four independent positive values are the real roots of

$$\begin{aligned} P_{\text{low}}(X) = & X^8 - 8X^7 + 12X^6 + 8X^5 - 22X^4 \\ & + 8X^3 + 12X^2 - 8X + 1. \end{aligned} \quad (37)$$

Zauner covariance and reciprocity supply the full lower stratum.

The forty-eight primitive characteristics contain four quadratic Fourier components and two conjugate pairs of quartic components. The two cyclic quartic quotient fields have relative equations

$$\begin{aligned} F_0/K_8: \quad X^4 - 6\phi X^2 + 3, \\ F_1/K_8: \quad X^4 + (6 - 6\phi)X^2 + 3. \end{aligned} \tag{38}$$

The exact models, fundamental-unit coordinates, and embeddings used below are fixed in `scripts/dimension_eight_quartic_bridge.gp`. Let η_0, η_1 be the anti-units selected there and let $\rho_1, \dots, \rho_4$ denote the four real embeddings in its local-Frobenius order. Define the oriented logarithmic resolvents

$$\begin{aligned} R_0 &= \log |\eta_0|_{\rho_3} + i \log |\eta_0|_{\rho_4}, \\ R_1 &= \log |\eta_1|_{\rho_2} + i \log |\eta_1|_{\rho_1}. \end{aligned} \tag{39}$$

In the PARI dual coordinates for the cyclic basis $C_4 \times C_2 \times C_2$ fixed by that script, Roblot's cyclic-quartic theorem proves the two modulus identities [10]

$$|L'_S(0, [1, 0, 0])| = |R_0|, \quad |L'_S(0, [1, 1, 0])| = |R_1|. \tag{40}$$

It does not prove the orientations.

The exact audit script `scripts/dimension_eight_exact_tcc.gp` constructs an algebraic candidate packet: its primitive squared-overlap field has degree 32 over $\mathbb{Q}$, all signed overlaps lie in one degree-64 field, and adjoining the Weyl phases gives a convention-selected degree-128 compositum. For each of the two formal shifts, the script verifies

$$\text{Tr } \Pi = 1, \quad \Pi^2 = \Pi, \quad \text{all } \binom{8}{2}^2 = 784 \text{ two-by-two minors vanish.} \tag{41}$$

Exact closure targets. The canonical analytic table would coincide with this candidate packet and therefore satisfy both formal shifts if the two convention-matched oriented identities

$$\begin{aligned} L'_S(0, [1, 0, 0]) &= R_0, \\ L'_S(0, [1, 1, 0]) &= R_1 \end{aligned} \tag{42}$$

hold. The lower-conductor special values and the algebraic calculation (41) are unconditional; their identification with the primitive analytic table is conditional on (42).

Rigorous double-sine windows locate all eight independent primitive analytic squares in the corresponding isolated algebraic intervals. At high precision the phase quotients in (42) equal $1 + O(10^{-114})$. This is strong evidence, not an equality proof: (40) permits an arbitrary unit-circle phase. The exact finite TCC orientation sieve selects one pair among the 64 natural algebraic orientations, but using that selection to deduce (42) would be circular. An oriented refinement of Roblot's theorem, or powered algebraicity for these two convention-matched cocycle values, would make (42) and hence dimension eight unconditional without any further finite computation.

8.3 Comparison

The two composite dimensions fail for different reasons:

d	unconditional analytic part	exact finite part	missing input
6	lower stratum, modulus and sign	225 minors	1 order-six value
8	15 lower overlaps and quartic moduli	2×784 minors	2 quartic orientations.

(43)

Dimension six is the smaller and conceptually sharper open Stark–Shintani identity. Dimension eight has the larger finite packet, but its remaining analytic obstruction is already separated into two explicit cyclic-quartic phases.

9 Scope and reproducibility

Theorem 13 proves the complete formal TCC in dimension five. Together with Section 2, it proves Theorem 1. It does not prove a dimension-five SIC, the TCC in arbitrary dimensions, or Zauner’s conjecture. The new number-theoretic step is Theorem 10: it proves the previously conjectural dimension-five ray packet by combining Kopp’s proved limit formula with the applicable case of Shintani’s proved algebraicity theorem and an effective height argument.

Section 8 records research-status closure targets, not conditional propositions masquerading as higher-dimensional theorems. It shows how far the proved low-dimensional machinery extends: the finite algebraic candidate packets pass exact cancellation audits, while their identification with the primitive analytic values still requires (33) in dimension six and the two orientations (42) in dimension eight. More decimal recognition cannot replace those missing special-value proofs.

The supplementary package contains:

- the 25-characteristic bridge certificate and generator;
- the exact C_8 Fourier-support audit excluding the quadratic character;
- the PARI transcript with `bnfcertify=1` and all positive ray-class logs;
- the Sturm root-isolation transcript;
- the exact interval/conjugate/factor embedding certificate;
- the formal 100-minor Laurent certificate; and
- the exact four-factor minor-reduction transcript;
- the Shintani sign-class, conductor, and exponent-5760 audit;
- the labeled K -isomorphism, unit, Sturm, and Frobenius-orbit certificate; and
- the rigorous Arb double-sine enclosure and height-gap transcript.
- the dimension-four 36-minor, ray, multiplier, and PARI certificates;
- the dimension-six ray, lower-stratum, Frobenius-orientation, and 225-minor audit scripts;
- the exact dimension-six fixed-point factorization (34);
- the dimension-eight conductor-lowering, lower-stratum, cyclic-quartic, Artin-label, and orientation audits; and
- the exact two-shift dimension-eight 784-minor audit script.

SHA-256 digests for the archived dimension-four and dimension-five artifacts are recorded in `certificates/SHA256SUMS`; the dimension-six and dimension-eight research scripts are fixed by the repository commit accompanying this manuscript.

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